Operating System Question Bank

Unit 3: Process Coordination

1. Explain producer-consumer problem.
2. Explain the following terms:
    i. Mutual Exclusion
    ii. Synchronization
    iii. Race condition
3. Write a deadlock-free solution for dining philosophers problem using semaphore.
4. What is critical section? Give semaphore solution for producer-consumer problem.
5. Explain with the concept of Inter-process Communication. Also discuss about pipes.
6. Explain the following functions with reference to semaphore programming in ‘C’: sem_post() and sem_wait().
7. Differentiate between counting semaphore and binary semaphore.
8. Explain critical section problem.
9. Discuss about the synchronization hardware.
10. What is the difference between critical section and critical region.
11. Write the structure of producer and consumer process in bounded buffer problem using semaphore and discuss how critical section requirements are fulfilled.
12. Explain any classical synchronization problem.
13. Write a semaphore solution for reader-writers problem.
14. Write a pseudo code for producer-consumer problem using semaphores.
15. Explain the semaphore usage and discuss its advantage and disadvantage.
16. Explain the necessary and sufficient conditions for the occurrence of a deadlock.
17. Explain with an appropriate example, how resource allocation graph determines a deadlock.
18. Write a short note on following:
    i. Resource allocation graph
    ii. Mutual exclusion
19. List the requirement of mutual exclusion.
20. List down the methods used for handling deadlocks and explain them.
21. What do you mean by deadlock prevention? What are the necessary conditions to prevent deadlock.
22. Explain hold and wait and circular wait condition with the help of resource allocation graph.
23. What is safe state? How safe state is used to avoid deadlock.
24. Explain Banker’s Algorithm.
25. Describe resource allocation graph algorithm.
26. Consider the following snapshot of a system:

	Allocation				Maximum				Available			
	A	B	C	D	A	B	C	D	A	B	C	D
P0	0	0	1	2	0	0	1	2	1	5	2	0
P1	1	0	0	0	1	7	5	0				

P2	1	3	5	4	2	3	5	6				
P3	0	6	3	2	0	6	5	2				
P4	0	0	1	4	0	6	5	6				

Answer the following questions using banker's algorithm.

i. What are the contents of Need matrix?

ii. Is the system in a safe state? What is safe sequence?

27. Explain busy waiting with appropriate example?
28. Consider the following snapshot of a system:

	Allocation			Maximum			Available		
	R1	R2	R3	R1	R2	R3	R1	R2	R3
P1	0	1	0	7	5	3	3	3	2
P2	2	0	0	3	2	2			
P3	3	0	2	9	0	2			
P4	2	1	1	2	2	2			
P5	0	0	2	4	3	3			

Answer the following questions using banker's algorithm.

i. What are the contents of Need matrix?

ii. Is the system in a safe state? What is safe sequence?

iii. If a request from process P1 arrives for (1, 0 2), can the request be granted immediately?

29. What are the various operations used to perform to recover from deadlock.
30. Explain deadlock detection methods.